

Assignment 3 COSC 2347

DUE 11:59 pm on 3/24/25 @ 11:59 pm. No late assignments.

Using correct C# style and naming conventions as detailed in the lectures and slides, create a C# OOP program as described below.

You were just hired by Huntsville's newest builder! You have 5 building types that you can build, and you've been asked to design a system that could track activities and information about your buildings.

(Residential Types)

Town Home

Single Family Home

(Commercial Types)

Warehouse

Office

Factory

Here are your design constraints:

1. Every building must be painted a color
2. Every building must have an address
3. All single-family homes are painted blue, townhouses are painted red.
4. All commercial buildings are painted grey.
5. All buildings have air conditioning, except warehouses
6. Toilets: town homes: 4, single family homes: 1, warehouses: 0, offices: 0, factories: 2
7. Doors: town homes: 1, single family homes: 2, warehouses: 4, offices: 4, factories: 6
8. Only residential buildings have sprinklers, no other building types have them

Design your classes as if you could build any of them, but you don't have to create code for classes you will not instantiate.

Your first orders have come in and the following events need to be handled:

Build a house

Build a warehouse

Build a town house

Turn on all buildings' AC systems

A fire breaks out at every building. Report what happened!

Print the status of each building. It should report everything you know about the building's status (instance variables).

Each of the buildings you create (instantiate) should be an object and the objects should react when you tell them what events are occurring to them!

Notes:

1. This program must be solved using OOP and reasonable class designs.
2. The point of this program is to exercise the information we've learned in chapters 6 and 7. If you solve the program using structured programming, you will receive 0 points.
3. Output the program's results to the console using the standard Console.WriteLine and Console.WriteLine commands.
4. Your program cannot throw exceptions from conversion results.
5. Capture the output screen with the correct results. If your output screen is disappearing before you can capture it, put in a ReadLine statement at the end of the program and then you can capture the output window. Now that we're not doing console redirection, the ReadLine statements will do what you expect and stop execution and wait.
6. Turn your complete source code along with the screen capture of your complete program output into Blackboard by the assignment due date. Failure to turn in full source code (in text format) and screen output snapshots showing all of your program output = grade of 0.
7. Like all assignments in this class, this is NOT a group project. Writing your code with someone else or sharing code is academic dishonesty and will be dealt with accordingly.
8. As explained in the syllabus, these instructions are subject to verbal change in this class. If you are not paying attention or not here and miss any changes in instructions, that is on you.

Here is sample output, but not THE full or correct output of the program. Follow the directions above.

Microsoft Visual Studio Debug Console

Built a new single family home!
Built a new factory!

Trying to turn on everyone's AC:
The single family home AC is on!
The factory AC is on!

There was a fire:
The single family home is on fire, but it has sprinklers!
The factory is on fire, but it doesn't have sprinklers!

Printing the building status:
single family home at 200 south broadway is blue and has an AC. toilets:1 doors:2. It has sprinklers.
factory at 105 main street is grey and has an AC. toilets:2 doors:6. It has no sprinklers.

C:\Users\rjg046\source\repos\Assignment 2.5\Assignment 2.5\bin\Debug\net8.0\Assignment 2.5.exe (process
th code 0.

To automatically close the console when debugging stops, enable Tools->Options->Debugging->Automaticall
le when debugging stops.

Press any key to close this window . . .